

Nộp bài tập homework ss09

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1. Why does skew occur during Clock Tree Synthesis (CTS)?

- Skew là khoảng chênh lệch thời gian clock của 2 flipflop khác nhau trong design.
- Việc xảy ra skew trong quá trình CTS có thể có nhiều lý do khác nhau ví dụ như:
 - Chiều dài dây khác nhau: Do chỉ có 1-2 path clock trong design nên việc phân bổ clock đến các cell phải có chiều dài dây khác nhau gây ra độ chênh lệch này.
 - Số lượng và kích thước buf/inv: khi phân bổ tín hiệu clock cho các cell thì trên đường đi sẽ được insert các buf để fix transistion, nên trên đường tới các cell càng dài thì sẽ insert càng nhiều buf, nên cũng gây ra skew
 - Ngoài ra còn có high fanout, RC delay cũng có thể gây ra skew.

2. Why do new setup or hold violations occur following the CTS step?

- Việc xuất hiện các path setup hoặc hold violate mới là do skew ở bước CTS này.
- Ở bước Placement, Clock là ideal (lý tưởng), được tools giả sử tín hiệu, nhưng khi đến bước CTS do các nguyên nhân kể ở bài 1 thì có thể xuất hiện thêm skew và latency nữa, những path đã MET ở Placement có thể vẫn bị violate ở bước CTS tùy thuộc vào đó là Positive hay Negative Skew sẽ ảnh hưởng đến Setup hoặc Hold Timing.

3. Source script của 02_place_opt.inv và 03_cts.tcl qua các target_skew khác nhau (0.2 ns, 0.35ns, 0.5ns)

3.1. Target_skew: 0.2ns

- Số lượng buffer trong report ccopt_clock_trees.rpt: 216 buf

Cell type	Count	Area	Capacitance
Buffers	216	2605.478	0.727
Inverters	1	5.443	0.003
Integrated Clock Gates	0	0.000	0.000
Non-Integrated Clock Gates	0	0.000	0.000
Clock Logic	3	28818.144	0.494
All	220	31429.066	1.223

- Report Setup/Hold timing, check if the skew matching the target?
 - Report 1 path timing setup của group reg2reg

Path 1: VIOLATED Setup Check with Pin i_croc_soc/i_croc/i_core_wrap/i_ibex_if_stage_i_prefetch_buffer_i_fifo_i_rdata_q_26__reg/CLK
 Endpoint: i_croc_soc/i_croc/i_core_wrap/i_ibex_if_stage_i_prefetch_buffer_i_fifo_i_rdata_q_26__reg/D (*) checked with leading edge of 'clk_sys'
 Beginpoint: i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_i_controller_i_instr_i_31__reg/Q (v) triggered by leading edge of 'clk_sys'
 Path Groups: {reg2reg}
 Analysis View: func_view_wc
 Other End Arrival Time 1.490
 - Setup 0.153
 + Phase Shift 10.000
 + CPFR Adjustment 0.017
 - Uncertainty 0.100
 = Required Time 11.254
 - Arrival Time 11.740
 = Slack Time -0.486
 Clock Rise Edge 0.000
 + Drive Adjustment 0.260
 = Beginpoint Arrival Time 0.260
 Timing Path:

Pin	Cell	Fanout	Load	Slew	Delay	Arrival Time	Pin Location
clk_i	-	1	0.245	0.348	-	0.260	(220.00, 1638.50)
pad_clk_i/pad	sg13g2_IOPadIn	1	0.245	0.348	0.002	0.262	(220.00, 1638.50)
pad_clk_i/p2c	sg13g2_IOPadIn	-	0.169	0.440	0.277	0.539	(220.03, 1460.14)
i_croc_soc/i_croc/CTS_ccl_a_buf_00491/A	sg13g2_buf_8	7	0.169	0.290	0.033	0.571	(593.76, 1142.40)
i_croc_soc/i_croc/CTS_ccl_a_buf_00491/X	sg13g2_buf_8	-	0.435	0.380	0.398	0.970	(591.84, 1143.24)
i_croc_soc/i_croc/i_core_wrap/CTS_ccl_a_buf_00370/A	sg13g2_buf_8	25	0.435	0.404	0.091	1.060	(1137.12, 888.72)
i_croc_soc/i_croc/i_core_wrap/CTS_ccl_a_buf_00370/X	sg13g2_buf_8	-	0.099	0.110	0.292	1.352	(1139.04, 887.88)
i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_i_controller_i_instr_i_31__reg/CLK	sg13g2_dfrbp_2	15	0.099	0.111	0.012	1.364	(1146.24, 727.02)
i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_i_controller_i_instr_i_31__reg/Q	sg13g2_dfrbp_2	-	0.044	0.135	0.455	1.819	(1152.96, 727.44)

Clock Rise Edge 0.000
 + Drive Adjustment 0.243
 = Beginpoint Arrival Time 0.243
 Other End Path:

Pin	Cell	Fanout	Load	Slew	Delay	Arrival Time	Pin Location
clk_i	-	1	0.195	0.337	-	0.243	(220.00, 1638.50)
pad_clk_i/pad	sg13g2_IOPadIn	1	0.195	0.337	0.001	0.244	(220.00, 1638.50)
pad_clk_i/p2c	sg13g2_IOPadIn	-	0.169	0.440	0.277	0.521	(220.03, 1460.14)
i_croc_soc/CTS_ccl_a_buf_00493/A	sg13g2_buf_8	7	0.169	0.289	0.015	0.536	(225.60, 1410.36)
i_croc_soc/CTS_ccl_a_buf_00493/X	sg13g2_buf_8	-	0.394	0.350	0.353	0.889	(227.52, 1409.52)
i_croc_soc/i_croc/i_core_wrap/CTS_ccl_a_buf_00374/A	sg13g2_buf_4	12	0.394	0.393	0.123	1.013	(1055.04, 540.96)
i_croc_soc/i_croc/i_core_wrap/CTS_ccl_a_buf_00374/X	sg13g2_buf_4	-	0.157	0.271	0.465	1.478	(1056.96, 540.12)
i_croc_soc/i_croc/i_core_wrap/i_ibex_if_stage_i_prefetch_buffer_i_fifo_i_rdata_q_26__reg/CLK	sg13g2_dfrbp_1	32	0.157	0.272	0.013	1.490	(1102.56, 522.48)

- Check skew dùng công thức: skew = capture – launch
- Ở path skew = 1.49 – 1.364 = 0.126 (Đạt target_skew)
- Skew này khá lớn nên violate setup timing!

- Report 1 path timing hold của group reg2reg

Path 1: MET Hold Check with Pin i_croc_soc/i_croc/i_gpio/gen_gpios_28__i_sync/serial_o_reg/CLK
 Endpoint: i_croc_soc/i_croc/i_gpio/gen_gpios_28__i_sync/serial_o_reg/D (v) checked with leading edge of 'clk_sys'
 Beginpoint: i_croc_soc/i_croc/i_gpio/gen_gpios_28__i_sync/reg_q_0__reg/Q (v) triggered by leading edge of 'clk_sys'
 Path Groups: {reg2reg}
 Analysis View: func_view_bc
 Other End Arrival Time 0.767
 + Hold 0.000
 + Phase Shift 0.000
 - CPFR Adjustment 0.007
 + Uncertainty 0.100
 = Required Time 0.860
 - Arrival Time 0.879
 = Slack Time 0.019
 Clock Rise Edge 0.000
 + Drive Adjustment 0.122
 = Beginpoint Arrival Time 0.122
 Timing Path:

Pin	Cell	Fanout	Load	Slew	Delay	Arrival Time	Pin Location
clk_i	-	1	0.196	0.167	-	0.122	(220.00, 1638.50)
pad_clk_i/pad	sg13g2_IOPadIn	1	0.196	0.167	0.000	0.122	(220.00, 1638.50)
pad_clk_i/p2c	sg13g2_IOPadIn	-	0.176	0.203	0.102	0.224	(220.03, 1460.14)
i_croc_soc/i_croc/CTS_ccl_a_buf_00487/A	sg13g2_buf_8	7	0.176	0.139	0.032	0.256	(410.40, 1233.12)
i_croc_soc/i_croc/CTS_ccl_a_buf_00487/X	sg13g2_buf_8	-	0.420	0.150	0.117	0.374	(408.48, 1233.96)
i_croc_soc/i_croc/CTS_ccl_a_buf_00298/A	sg13g2_buf_4	26	0.420	0.219	0.091	0.465	(361.44, 643.44)
i_croc_soc/i_croc/CTS_ccl_a_buf_00298/X	sg13g2_buf_4	-	0.016	0.032	0.127	0.592	(363.36, 644.28)
i_croc_soc/i_croc/CTS_csk_buf_00731/A	sg13g2_buf_8	1	0.016	0.032	0.002	0.594	(363.84, 673.68)
i_croc_soc/i_croc/CTS_csk_buf_00731/X	sg13g2_buf_8	-	0.178	0.070	0.084	0.678	(361.92, 674.52)
i_croc_soc/i_croc/i_gpio/gen_gpios_28__i_sync/reg_q_0__reg/CLK	sg13g2_dfrbp_1	32	0.178	0.072	0.013	0.691	(208.32, 696.36)
i_croc_soc/i_croc/i_gpio/gen_gpios_28__i_sync/reg_q_0__reg/Q	sg13g2_dfrbp_1	-	0.024	0.057	0.185	0.877	(217.92, 697.20)
i_croc_soc/i_croc/i_gpio/gen_gpios_28__i_sync/serial_o_reg/D	sg13g2_dfrbp_1	1	0.024	0.058	0.002	0.879	(312.48, 764.82)

Clock Rise Edge	0.000							
+ Drive Adjustment	0.129							
= Beginpoint Arrival Time	0.129							
Other End Path:								
Pin	Cell	Fanout	Load	Slew	Delay	Arrival Time	Pin Location	
clk_i	-	1	0.240	0.171	-	0.129	(220.00, 1638.50)	
pad_clk_i/pad	sg13g2_IOPadIn	1	0.240	0.171	0.000	0.129	(220.00, 1638.50)	
pad_clk_i/p2c	sg13g2_IOPadIn	-	0.176	0.203	0.102	0.231	(220.03, 1460.14)	
i_croc_soc/CTS_ccl_a_buf_00487/A	sg13g2_buf_8	7	0.176	0.139	0.032	0.263	(410.40, 1233.12)	
i_croc_soc/CTS_ccl_a_buf_00487/X	sg13g2_buf_8	-	0.420	0.150	0.117	0.381	(408.48, 1233.96)	
i_croc_soc/i_croc/CTS_ccl_a_buf_00254/A	sg13g2_buf_8	26	0.420	0.219	0.091	0.472	(373.44, 688.80)	
i_croc_soc/i_croc/CTS_ccl_a_buf_00254/X	sg13g2_buf_8	-	0.022	0.025	0.103	0.574	(371.52, 689.64)	
i_croc_soc/i_croc/CTS_cdb_buf_00713/A	sg13g2_buf_2	2	0.022	0.025	0.001	0.576	(370.08, 741.72)	
i_croc_soc/i_croc/CTS_cdb_buf_00713/X	sg13g2_buf_2	-	0.157	0.216	0.185	0.760	(369.12, 742.56)	
i_croc_soc/i_croc/i_gpio/gen_gpios_28_i_sync/serial_o_reg/CLK	sg13g2_dfrbp_1	31	0.157	0.216	0.007	0.767	(314.40, 764.40)	

- Ở path hold này, skew = $0.767 - 0.691 = 0.076$ (Đạt target_skew)
- Skew ở path này rất nhỏ nên MET timing!

3.2. Target_skew: 0.35ns

- Số lượng buffer trong report ccopt_clock_trees.rpt: 178 buf (nhỏ hơn 0.2ns là 216 buf)

Cell type	Count	Area	Capacitance
Buffers	178	2431.296	0.704
Inverters	1	5.443	0.003
Integrated Clock Gates	0	0.000	0.000
Non-Integrated Clock Gates	0	0.000	0.000
Clock Logic	3	28818.144	0.494
All	182	31254.883	1.201

- Report Setup/Hold timing, check if the skew matching the target?
 - Report 1 path timing setup của group reg2reg

Path 1: VIOLATED Setup Check with Pin i_croc_soc/i_croc/i_core_wrap/i_ibex_if_stage_1_prefetch_buffer_1_fifo_1_rdata_q_13_reg/CLK								
Endpoint: i_croc_soc/i_croc/i_core_wrap/i_ibex_if_stage_1_prefetch_buffer_1_fifo_1_rdata_q_13_reg/0 (*) checked with leading edge of 'clk_sys'								
Beginpoint: i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_31_reg/Q (v) triggered by leading edge of 'clk_sys'								
Path Groups: (reg2reg)								
Analysis View: Func_View_wc								
Other End Arrival Time								
- Setup	1.461							
- Phase Shift	0.175							
+ CPFR Adjustment	10.000							
- Uncertainty	0.017							
= Required Time	0.100							
= Arrival Time	11.203							
= Slack Time	11.868							
= Slack Time	-0.666							
Clock Rise Edge								
+ Drive Adjustment	0.000							
= Beginpoint Arrival Time	0.260							
Timing Path:								
Pin	Cell	Fanout	Load	Slew	Delay	Arrival Time	Pin Location	
clk_i	-	1	0.245	0.348	-	0.260	(220.00, 1638.50)	
pad_clk_i/pad	sg13g2_IOPadIn	1	0.245	0.348	0.002	0.262	(220.00, 1638.50)	
pad_clk_i/p2c	sg13g2_IOPadIn	-	0.050	0.161	0.161	0.423	(220.03, 1460.14)	
i_croc_soc/CTS_ccl_buf_01256/A	sg13g2_buf_8	1	0.050	0.105	0.004	0.427	(325.44, 1304.52)	
i_croc_soc/CTS_ccl_buf_01256/X	sg13g2_buf_8	-	0.277	0.253	0.241	0.607	(327.36, 1303.68)	
i_croc_soc/CTS_ccl_a_buf_01252/A	sg13g2_buf_8	6	0.277	0.275	0.080	0.747	(1104.96, 832.44)	
i_croc_soc/CTS_ccl_a_buf_01252/X	sg13g2_buf_8	-	0.357	0.313	0.369	1.116	(1103.04, 833.28)	
i_croc_soc/i_croc/i_core_wrap/CTS_ccl_a_buf_01129/A	sg13g2_buf_8	26	0.357	0.322	0.039	1.155	(1113.60, 817.32)	
i_croc_soc/i_croc/i_core_wrap/CTS_ccl_a_buf_01129/X	sg13g2_buf_8	-	0.177	0.162	0.316	1.471	(1111.68, 818.16)	
i_croc_soc/i_croc/i_core_wrap/CTS_ibex_id_stage_1_controller_1_instr_1_31_reg/CLK	sg13g2_dfrbp_2	32	0.177	0.162	0.010	1.482	(1146.24, 727.02)	
i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_31_reg/Q	sg13g2_dfrbp_2	-	0.044	0.133	0.480	1.961	(1152.96, 727.44)	
i_croc_soc/i_croc/i_core_wrap/FE_RC_193_0/A	sg13g2_nor2_2	6	0.044	0.134	0.007	1.968	(1066.08, 726.60)	
i_croc_soc/i_croc/i_core_wrap/FE_RC_193_0/Y	sg13g2_nor2_2	-	0.007	0.092	0.131	2.098	(1066.08, 727.44)	
i_croc_soc/i_croc/i_core_wrap/FE_OCP6236_FE_BN_06_0/A	sg13g2_inv_2	1	0.007	0.002	0.001	2.099	(1061.76, 729.06)	

Clock Rise Edge	0.000							
+ Drive Adjustment	0.243							
= Beginpoint Arrival Time	0.243							
Other End Path:								
Pin	Cell	Fanout	Load	Slew	Delay	Arrival Time	Pin Location	
clk_i	-	1	0.195	0.337	-	0.243	(220.00, 1638.50)	
pad_clk_i/pad	sg13g2_IOPadIn	1	0.195	0.337	0.001	0.244	(220.00, 1638.50)	
pad_clk_i/p2c	sg13g2_IOPadIn	-	0.050	0.160	0.161	0.405	(220.03, 1460.14)	
i_croc_soc/CTS_ccl_buf_01256/A	sg13g2_buf_8	1	0.050	0.105	0.004	0.409	(325.44, 1304.52)	
i_croc_soc/CTS_ccl_buf_01256/X	sg13g2_buf_8	-	0.277	0.253	0.241	0.649	(327.36, 1303.68)	
i_croc_soc/CTS_ccl_a_buf_01252/A	sg13g2_buf_8	6	0.277	0.275	0.080	0.729	(1104.96, 832.44)	
i_croc_soc/CTS_ccl_a_buf_01252/X	sg13g2_buf_8	-	0.357	0.313	0.369	1.098	(1103.04, 833.28)	
i_croc_soc/i_croc/i_core_wrap/CTS_ccl_a_buf_01171/A	sg13g2_buf_8	26	0.357	0.323	0.046	1.145	(1115.52, 877.04)	
i_croc_soc/i_croc/i_core_wrap/CTS_ccl_a_buf_01171/X	sg13g2_buf_8	-	0.153	0.148	0.293	1.438	(1117.44, 876.20)	
i_croc_soc/i_croc/i_core_wrap/i_ibex_if_stage_1_prefetch_buffer_1_fifo_1_rdata_q_13_reg/CLK	sg13g2_dfrbp_1	32	0.153	0.152	0.023	1.461	(1084.32, 548.52)	

- Check skew dùng công thức: $\text{skew} = \text{capture} - \text{launch}$
- Ở path skew = $1.461 - 1.482 = -0.021$ (Đạt target_skew)
- Dấu âm là vì skew âm, tệ cho setup, chứ so sánh target_skew thì $|-0.021| < 0.35$

○ Report 1 path timing hold của group reg2reg

Path 1: MET Hold Check with Pin i_croc_soc/i_croc/i_timer/s_ref_clk3_reg/CLK
 Endpoint: i_croc_soc/i_croc/i_timer/s_ref_clk3_reg/D (v) checked with leading edge of 'clk_sys'
 Beginpoint: i_croc_soc/i_croc/i_timer/s_ref_clk2_reg/Q (v) triggered by leading edge of 'clk_sys'
 Path Groups: {reg2reg}
 Analysis View: func_view_bc
 Other End Arrival Time 0.764
 + Hold 0.009
 + Phase Shift 0.000
 - CPPR Adjustment 0.007
 + Uncertainty 0.100
 = Required Time 0.866
 Arrival Time 0.891
 Slack Time 0.025
 Clock Rise Edge 0.000
 + Drive Adjustment 0.122
 = Beginpoint Arrival Time 0.122
 Timing Path:

Pin	Cell	Fanout	Load	Slew	Delay	Arrival Time	Pin Location
clk_i	-	1	0.196	0.167	-	0.122	(220.00, 1638.50)
pad_clk_i/pad	sg13g2_IOPadIn	1	0.196	0.167	0.000	0.122	(220.00, 1638.50)
pad_clk_i/p2c	sg13g2_IOPadIn	-	0.051	0.070	0.065	0.186	(220.03, 1460.14)
i_croc_soc/CTS_ccl_buf_01256/A	sg13g2_buf_8	1	0.051	0.046	0.004	0.190	(325.44, 1304.52)
i_croc_soc/CTS_ccl_buf_01256/X	sg13g2_buf_8	-	0.284	0.100	0.074	0.265	(327.36, 1303.68)
i_croc_soc/CTS_ccl_a_buf_01246/A	sg13g2_buf_8	6	0.284	0.159	0.070	0.335	(657.12, 1115.52)
i_croc_soc/CTS_ccl_a_buf_01246/X	sg13g2_buf_8	-	0.378	0.137	0.124	0.459	(659.04, 1114.68)
i_croc_soc/i_croc/CTS_ccl_a_buf_01039/A	sg13g2_buf_4	23	0.378	0.182	0.050	0.509	(506.88, 892.92)
i_croc_soc/i_croc/CTS_ccl_a_buf_01039/X	sg13g2_buf_4	-	0.171	0.125	0.193	0.701	(508.80, 893.76)
i_croc_soc/i_croc/i_timer/s_ref_clk2_reg/CLK	sg13g2_dfrbp_1	32	0.171	0.126	0.009	0.710	(521.28, 885.36)
i_croc_soc/i_croc/i_timer/s_ref_clk2_reg/Q	sg13g2_dfrbp_1	-	0.010	0.028	0.180	0.890	(511.68, 886.20)
i_croc_soc/i_croc/i_timer/s_ref_clk3_reg/D	sg13g2_dfrbp_1	2	0.010	0.028	0.001	0.891	(509.76, 895.86)

Clock Rise Edge 0.000
 + Drive Adjustment 0.129
 = Beginpoint Arrival Time 0.129
 Other End Path:

Pin	Cell	Fanout	Load	Slew	Delay	Arrival Time	Pin Location
clk_i	-	1	0.240	0.171	-	0.129	(220.00, 1638.50)
pad_clk_i/pad	sg13g2_IOPadIn	1	0.240	0.171	0.000	0.129	(220.00, 1638.50)
pad_clk_i/p2c	sg13g2_IOPadIn	-	0.051	0.070	0.065	0.193	(220.03, 1460.14)
i_croc_soc/CTS_ccl_buf_01256/A	sg13g2_buf_8	1	0.051	0.046	0.004	0.197	(325.44, 1304.52)
i_croc_soc/CTS_ccl_buf_01256/X	sg13g2_buf_8	-	0.284	0.100	0.074	0.271	(327.36, 1303.68)
i_croc_soc/CTS_ccl_a_buf_01246/A	sg13g2_buf_8	6	0.284	0.159	0.070	0.342	(657.12, 1115.52)
i_croc_soc/CTS_ccl_a_buf_01246/X	sg13g2_buf_8	-	0.378	0.137	0.124	0.465	(659.04, 1114.68)
i_croc_soc/i_croc/CTS_ccl_a_buf_00953/A	sg13g2_buf_2	23	0.378	0.182	0.050	0.516	(534.24, 915.60)
i_croc_soc/i_croc/CTS_ccl_a_buf_00953/X	sg13g2_buf_2	-	0.165	0.228	0.238	0.754	(535.20, 916.44)
i_croc_soc/i_croc/i_timer/s_ref_clk3_reg/CLK	sg13g2_dfrbp_1	32	0.165	0.229	0.010	0.764	(511.68, 896.28)

- Ở path hold này, skew = $0.764 - 0.710 = 0.054$ (Đạt target_skew)
- Skew ở path này rất nhỏ nên MET timing!

3.3. Target_skew: 0.5ns

- Số lượng buffer trong report ccopt_clock_trees.rpt: 177 buf (nhỏ hơn 0.35ns là 178 buf)

Cell type	Count	Area	Capacitance
Buffers	177	2168.208	0.583
Inverters	1	5.443	0.003
Integrated Clock Gates	0	0.000	0.000
Non-Integrated Clock Gates	0	0.000	0.000
Clock Logic	3	28818.144	0.494
All	181	30991.795	1.079

- Report Setup/Hold timing, check if the skew matching the target?
 - Report 1 path timing setup của group reg2reg

Path 1: VIOLATED Setup Check with Pin i_croc_soc/i_croc/i_core_wrap/i_ibex_crash_dump_o_84__reg/CLK
 Endpoint: i_croc_soc/i_croc/i_core_wrap/i_ibex_crash_dump_o_84__reg/D (*) checked with leading edge of 'clk_sys'
 Beginpoint: i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_i_controller_i_instr_1_31__reg/Q (v) triggered by leading edge of 'clk_sys'
 Path Groups: {reg2reg}
 Analysis View: func_view_wc

Other End Arrival Time 1.447
 - Setup 0.189
 + Phase Shift 10.000
 + CPPR Adjustment 0.017
 - Uncertainty 0.100
 = Required Time 11.175
 - Arrival Time 11.941
 - Slack Time -0.766

Clock Rise Edge 0.000
 + Drive Adjustment 0.260
 = Beginpoint Arrival Time 0.260

Timing Path:

Pin	Cell	Fanout	Load	Slew	Delay	Arrival Time	Pin Location
clk_i	-	1	0.245	0.348	-	0.260	(220.00, 1638.50)
pad_clk_i/pad	sg13g2_IOPadIn	1	0.245	0.348	0.002	0.262	(220.00, 1638.50)
pad_clk_i/p2c	sg13g2_IOPadIn	-	0.009	0.072	0.111	0.373	(220.03, 1460.14)
i_croc_soc/CTS_ccl_buf_01850/A	sg13g2_buf_8	1	0.009	0.047	0.000	0.373	(220.80, 1455.72)
i_croc_soc/CTS_ccl_buf_01850/X	sg13g2_buf_8	-	0.312	0.282	0.235	0.608	(222.72, 1454.88)
i_croc_soc/CTS_ccl_a_buf_01846/A	sg13g2_buf_8	6	0.312	0.297	0.074	0.682	(1035.84, 843.36)
i_croc_soc/CTS_ccl_a_buf_01846/X	sg13g2_buf_8	-	0.344	0.299	0.378	1.061	(1037.76, 842.52)
i_croc_soc/i_croc/i_core_wrap/CTS_ccl_a_buf_01723/A	sg13g2_buf_4	26	0.344	0.305	0.032	1.092	(1115.52, 771.96)
i_croc_soc/i_croc/i_core_wrap/CTS_ccl_a_buf_01723/X	sg13g2_buf_4	-	0.171	0.292	0.446	1.538	(1117.44, 772.80)
i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_i_controller_i_instr_1_31__reg/CLK	sg13g2_dfrbp_2	32	0.171	0.293	0.009	1.547	(1146.24, 727.02)
i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_i_controller_i_instr_1_31__reg/Q	sg13g2_dfrbp_2	-	0.044	0.133	0.527	2.074	(1152.96, 727.44)

Clock Rise Edge 0.000
 + Drive Adjustment 0.243
 = Beginpoint Arrival Time 0.243

Other End Path:

Pin	Cell	Fanout	Load	Slew	Delay	Arrival Time	Pin Location
clk_i	-	1	0.195	0.337	-	0.243	(220.00, 1638.50)
pad_clk_i/pad	sg13g2_IOPadIn	1	0.195	0.337	0.001	0.244	(220.00, 1638.50)
pad_clk_i/p2c	sg13g2_IOPadIn	-	0.009	0.071	0.111	0.355	(220.03, 1460.14)
i_croc_soc/CTS_ccl_buf_01850/A	sg13g2_buf_8	1	0.009	0.047	0.000	0.355	(220.80, 1455.72)
i_croc_soc/CTS_ccl_buf_01850/X	sg13g2_buf_8	-	0.312	0.282	0.235	0.590	(222.72, 1454.88)
i_croc_soc/i_croc/CTS_ccl_a_buf_01838/A	sg13g2_buf_8	6	0.312	0.296	0.055	0.646	(508.80, 1127.28)
i_croc_soc/i_croc/CTS_ccl_a_buf_01838/X	sg13g2_buf_8	-	0.313	0.271	0.369	1.015	(506.88, 1128.12)
i_croc_soc/i_croc/i_core_wrap/CTS_ccl_a_buf_01559/A	sg13g2_buf_2	30	0.313	0.275	0.038	1.052	(812.16, 953.40)
i_croc_soc/i_croc/i_core_wrap/CTS_ccl_a_buf_01559/X	sg13g2_buf_2	-	0.083	0.279	0.386	1.438	(813.12, 954.24)
i_croc_soc/i_croc/i_core_wrap/i_ibex_crash_dump_o_84__reg/CLK	sg13g2_dfrbp_1	17	0.083	0.279	0.009	1.447	(871.20, 926.52)

- Check skew dùng công thức: skew = capture – launch
- Ở path skew = 1.447 – 1.547 = - 0.1 (Đạt target_skew)
- Dấu âm là vì skew âm, tệ cho setup, chứ so sánh target_skew thì $|-0.1| < 0.5$

○ Report 1 path timing hold của group reg2reg

Path 1: MET Hold Check with Pin i_croc_soc/i_croc/i_gpio/i_reg_file_reg_q_169__reg/CLK							
Endpoint: i_croc_soc/i_croc/i_gpio/i_reg_file_reg_q_169__reg/D (v) checked with leading edge of 'clk_sys'							
Beginpoint: i_croc_soc/i_croc/i_gpio/i_reg_file_new_reg_233__reg/Q_N (v) triggered by leading edge of 'clk_sys'							
Path Groups: {reg2reg}							
Analysis View: func_view_bc							
Other End Arrival Time			0.724				
+ Hold			0.012				
+ Phase Shift			0.000				
- CPDR Adjustment			0.007				
+ Uncertainty			0.100				
= Required Time			0.829				
Arrival Time			0.857				
Slack Time			0.028				
Clock Rise Edge			0.000				
+ Drive Adjustment			0.122				
= Beginpoint Arrival Time			0.122				
Timing Path:							
Pin	Cell	Fanout	Load	Slew	Delay	Arrival Time	Pin Location
clk_i	-	1	0.196	0.167	-	0.122	(220.00, 1638.50)
pad_clk_i/pad	sg13g2_IOPadIn	1	0.196	0.167	0.000	0.122	(220.00, 1638.50)
pad_clk_i/p2c	sg13g2_IOPadIn	-	0.010	0.032	0.045	0.167	(220.03, 1460.14)
i_croc_soc/CTS_ccl_buf_01850/A	sg13g2_buf_8	1	0.010	0.021	0.000	0.167	(220.80, 1455.72)
i_croc_soc/CTS_ccl_buf_01850/X	sg13g2_buf_8	-	0.319	0.117	0.072	0.240	(222.72, 1454.88)
i_croc_soc/CTS_ccl_a_buf_01840/A	sg13g2_buf_8	6	0.319	0.160	0.061	0.300	(508.32, 911.40)
i_croc_soc/CTS_ccl_a_buf_01840/X	sg13g2_buf_8	-	0.328	0.119	0.122	0.422	(510.24, 910.56)
i_croc_soc/i_croc/CTS_ccl_a_buf_01549/A	sg13g2_buf_4	23	0.328	0.142	0.020	0.442	(499.20, 896.28)
i_croc_soc/i_croc/CTS_ccl_a_buf_01549/X	sg13g2_buf_4	-	0.170	0.125	0.177	0.619	(497.28, 895.44)
i_croc_soc/i_croc/i_gpio/i_reg_file_new_reg_233__reg/CLK	sg13g2_dfrbp_1	32	0.170	0.127	0.014	0.634	(433.92, 866.04)
i_croc_soc/i_croc/i_gpio/i_reg_file_new_reg_233__reg/Q_N	sg13g2_dfrbp_1	-	0.008	0.038	0.160	0.794	(441.60, 866.46)
i_croc_soc/i_croc/i_gpio/_1094_/C	sg13g2_and3_1	2	0.008	0.038	0.000	0.794	(442.08, 873.60)
i_croc_soc/i_croc/i_gpio/_1094_/X	sg13g2_and3_1	-	0.004	0.017	0.063	0.857	(441.12, 873.18)
i_croc_soc/i_croc/i_gpio/i_reg_file_reg_q_169__reg/D	sg13g2_dfrbp_1	1	0.004	0.017	0.000	0.857	(429.60, 870.66)
Clock Rise Edge 0.000							
+ Drive Adjustment 0.129							
= Beginpoint Arrival Time 0.129							
Other End Path:							
Pin	Cell	Fanout	Load	Slew	Delay	Arrival Time	Pin Location
clk_i	-	1	0.240	0.171	-	0.129	(220.00, 1638.50)
pad_clk_i/pad	sg13g2_IOPadIn	1	0.240	0.171	0.000	0.129	(220.00, 1638.50)
pad_clk_i/p2c	sg13g2_IOPadIn	-	0.010	0.032	0.045	0.174	(220.03, 1460.14)
i_croc_soc/CTS_ccl_buf_01850/A	sg13g2_buf_8	1	0.010	0.021	0.000	0.175	(220.80, 1455.72)
i_croc_soc/CTS_ccl_buf_01850/X	sg13g2_buf_8	-	0.319	0.117	0.072	0.247	(222.72, 1454.88)
i_croc_soc/i_croc/CTS_ccl_a_buf_01844/A	sg13g2_buf_8	6	0.319	0.160	0.060	0.307	(493.92, 1074.36)
i_croc_soc/i_croc/CTS_ccl_a_buf_01844/X	sg13g2_buf_8	-	0.335	0.109	0.115	0.422	(492.00, 1075.20)
i_croc_soc/i_croc/CTS_ccl_a_buf_01691/A	sg13g2_buf_2	25	0.335	0.183	0.053	0.475	(430.56, 941.64)
i_croc_soc/i_croc/CTS_ccl_a_buf_01691/X	sg13g2_buf_2	-	0.163	0.227	0.233	0.709	(429.60, 940.80)
i_croc_soc/i_croc/i_gpio/i_reg_file_reg_q_169__reg/CLK	sg13g2_dfrbp_1	32	0.163	0.228	0.015	0.724	(431.52, 870.24)

- Ở path hold này, skew = $0.724 - 0.634 = 0.09$ (Đạt target_skew)
- Skew ở path này rất nhỏ nên MET timing!